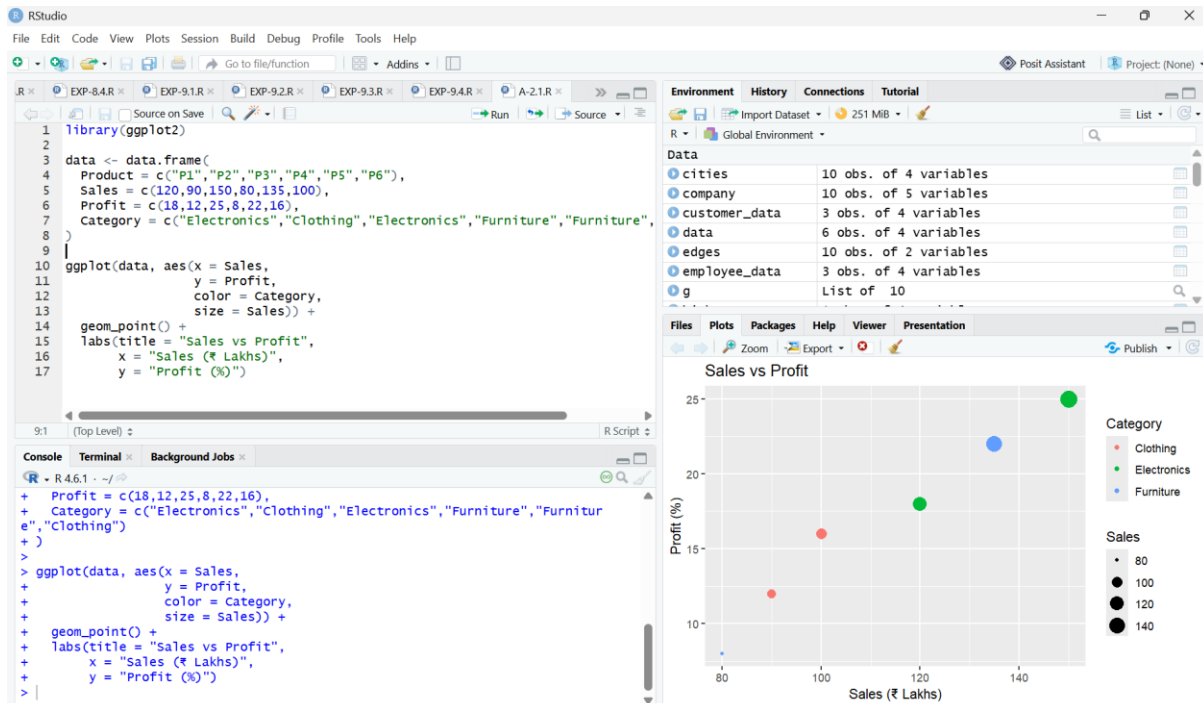


CO1_AT2

Question 1: Mapping Data onto Aesthetics



CODE:

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  Product = c("P1", "P2", "P3", "P4", "P5", "P6"),
```

```
  Sales = c(120, 90, 150, 80, 135, 100),
```

```
  Profit = c(18, 12, 25, 8, 22, 16),
```

```
  Category = c("Electronics", "Clothing", "Electronics", "Furniture", "Furniture", "Clothing")
```

```
)
```

```
ggplot(data, aes(x = Sales,
```

```
  y = Profit,
```

```
  color = Category,
```

```
  size = Sales)) +
```

```
geom_point() +
```

```
labs(title = "Sales vs Profit",
```

x = "Sales (₹ Lakhs)",

y = "Profit (%)"

2. Aesthetic Mapping

- **Category → Color**
- **Sales → Size**

3. Interpretation

- Sales and Profit show a **positive relationship**.
- Products with higher sales generally have higher profit.
- Electronics products show the highest sales and profit values.
- Furniture and Clothing have comparatively lower values.

4. Type of Scale

Variable Scale Type

Sales Continuous

Profit Continuous

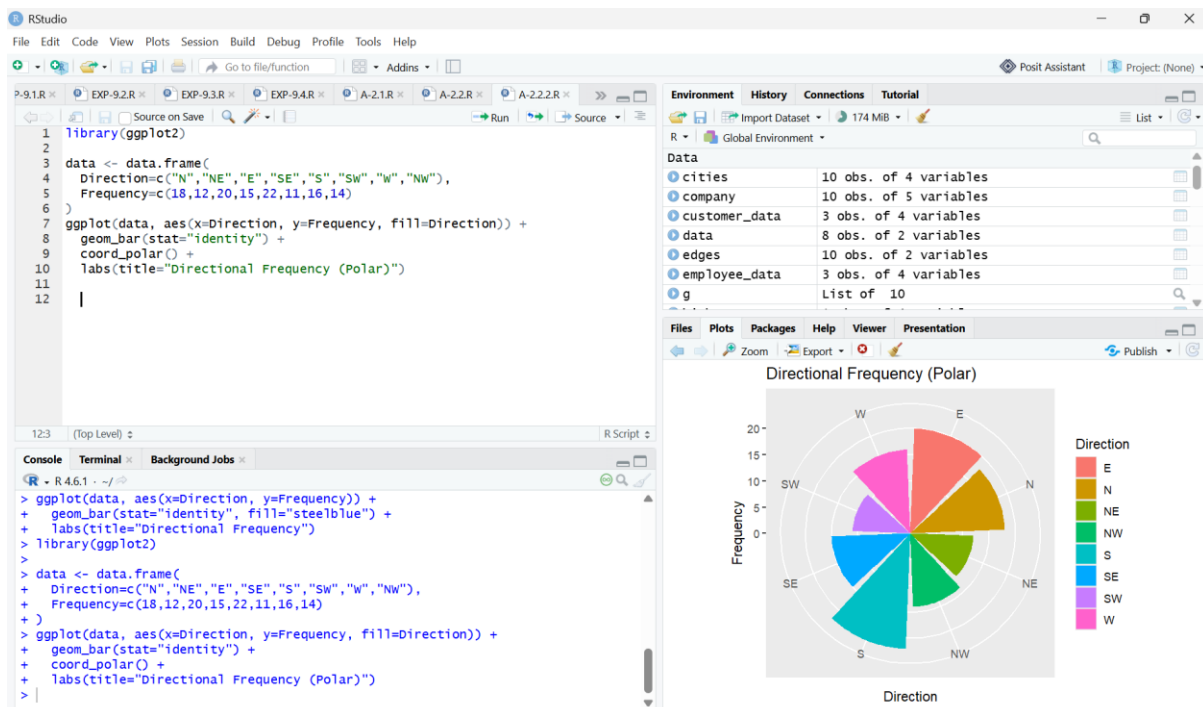
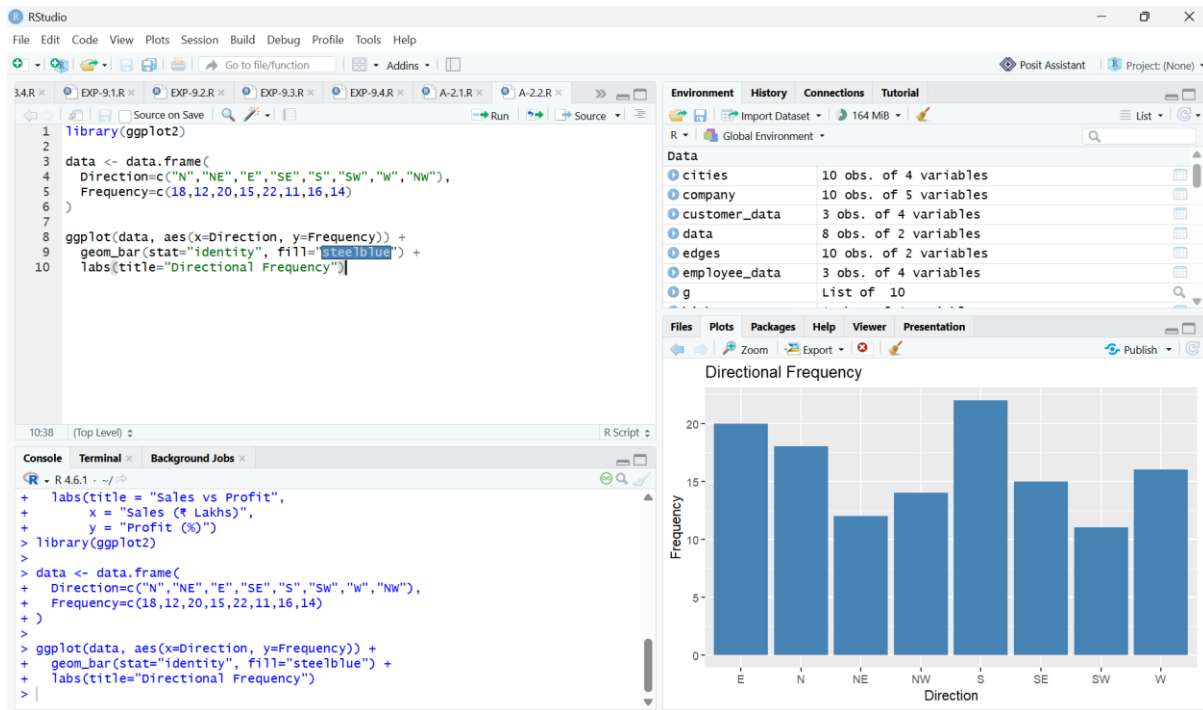
Category Categorical (Discrete)

5. Alternative Visualization

Bubble Chart (or Scatter Plot with Labels)

It can display Sales, Profit, and Category more clearly while emphasizing larger sales.

Question 2: Coordinate Systems



CODE:

```
library(ggplot2)
```

```
data <- data.frame(
```

```

Direction=c("N","NE","E","SE","S","SW","W","NW"),
Frequency=c(18,12,20,15,22,11,16,14)
)

ggplot(data, aes(x=Direction, y=Frequency)) +
  geom_bar(stat="identity", fill="steelblue") +
  labs(title="Directional Frequency")

ggplot(data, aes(x=Direction, y=Frequency, fill=Direction)) +
  geom_bar(stat="identity") +
  coord_polar() +
  labs(title="Directional Frequency (Polar)")

```

3. Comparison

Cartesian

- Easier to compare frequencies accurately.
- Better for numerical comparison.

Polar

- Better represents circular or directional data.
- Provides intuitive visualization of compass directions.

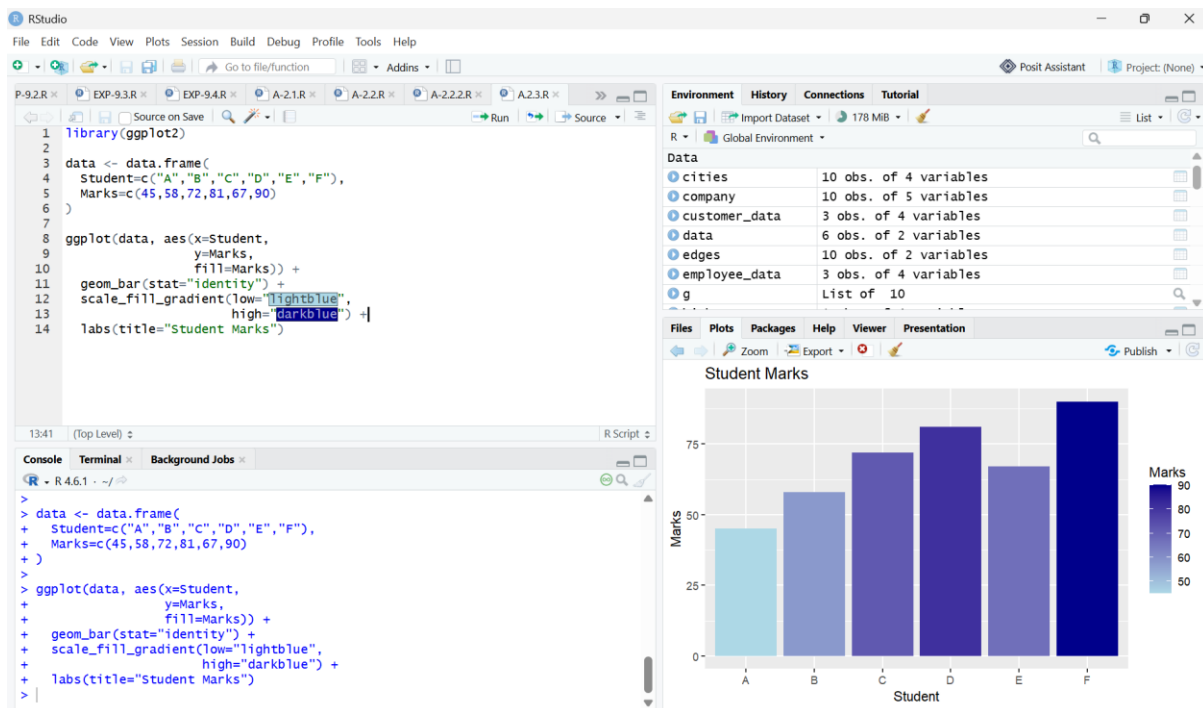
4. Appropriate Coordinate System

Polar Coordinate System

5. Justification

Since the data represents **compass directions**, the polar coordinate system naturally reflects the circular arrangement of directions and improves interpretation.

Question 3: Color Scales



CODE:

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  Student=c("A","B","C","D","E","F"),
```

```
  Marks=c(45,58,72,81,67,90)
```

```
)
```

```
ggplot(data, aes(x=Student,
```

```
  y=Marks,
```

```
  fill=Marks)) +
```

```
geom_bar(stat="identity") +
```

```
scale_fill_gradient(low="lightblue",
```

```
  high="darkblue") +
```

```
labs(title="Student Marks")
```

2. Highest and Lowest Marks

- Highest: Student F (90)
- Lowest: Student A (45)

3. Why Sequential Color Scale?

A sequential color scale is suitable because Marks are continuous numerical values, and increasing color intensity represents increasing marks.

4. Color-blind Friendly Palette

Viridis

Example:

```
scale_fill_viridis_c()
```

5. Sequential vs Categorical Color Scale

Sequential

Categorical

Used for continuous data

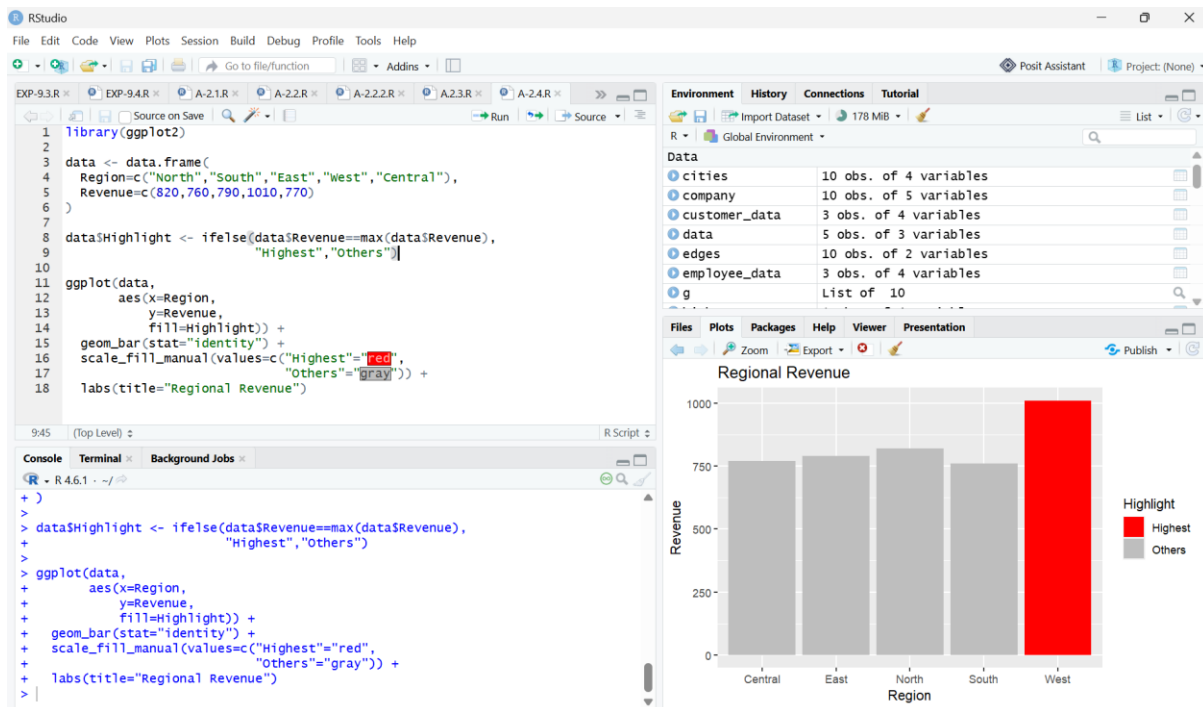
Used for categories

Color intensity changes gradually Different colors represent different groups

Example: Marks

Example: Gender or Department

Question 4: Highlighting with Color



CODE:

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  Region=c("North","South","East","West","Central"),
```

```
  Revenue=c(820,760,790,1010,770)
```

```
)
```

```
data$Highlight <- ifelse(data$Revenue==max(data$Revenue),
```

```
  "Highest","Others")
```

```
ggplot(data,
```

```
  aes(x=Region,
```

```
    y=Revenue,
```

```
    fill=Highlight)) +
```

```
geom_bar(stat="identity") +
```

```
scale_fill_manual(values=c("Highest"="red",
```

```
  "Others"="gray")) +
```

```
labs(title="Regional Revenue")
```

2. How Highlighting Improves Interpretation

- Draws immediate attention to the most important data point.
- Makes comparisons easier.
- Helps identify trends quickly.

3. Highest Performing Region

West (₹1010 Crores)

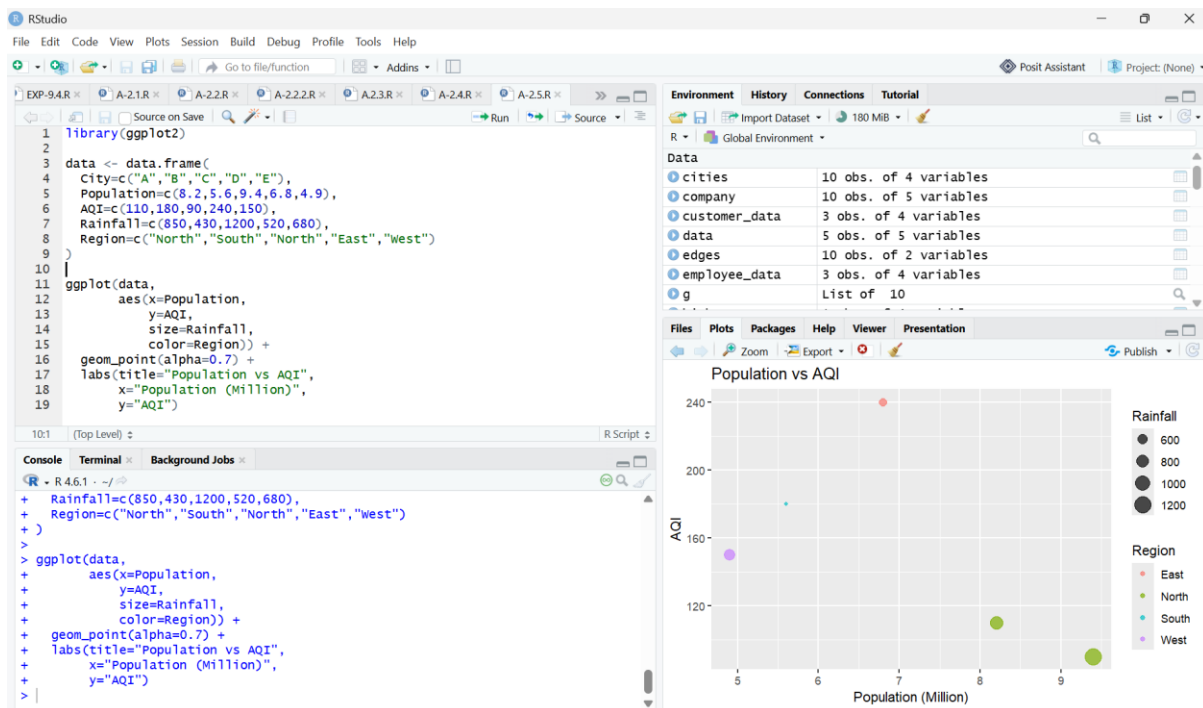
4. Effect of Excessive Colors

- Makes charts confusing.
- Reduces readability.
- Distracts viewers.
- Hides important information.

5. Alternative Highlighting Technique

- Labels
- Bold borders
- Different bar pattern (texture)
- Annotation

Question 5: Integrated Visualization



CODE:

```
library(ggplot2)

data <- data.frame(

  City=c("A","B","C","D","E"),

  Population=c(8.2,5.6,9.4,6.8,4.9),

  AQI=c(110,180,90,240,150),

  Rainfall=c(850,430,1200,520,680),

  Region=c("North","South","North","East","West")

)

ggplot(data,

  aes(x=Population,

    y=AQI,

    size=Rainfall,

    color=Region)) +

  geom_point(alpha=0.7) +

  labs(title="Population vs AQI",
```

```
x="Population (Million)",  
y="AQI")
```

2. Aesthetic Mapping

Variable Aesthetic

Population X-axis

AQI Y-axis

Rainfall Bubble Size

Region Color

3. Interpretation

- Cities with higher population generally tend to have higher AQI.
- City D has the highest AQI despite a moderate population.
- Rainfall varies across cities and is represented by bubble size.

4. Continuous and Categorical Variables

Continuous Variables

- Population
- AQI
- Rainfall

Categorical Variable

- Region

5. Improvement for Better Readability

- Add city labels using `geom_text()`.
- Use a color-blind-friendly palette such as **Viridis**.